NMEC595 – Research Methodology

Lecture 14: Comparing Two Population Parameters

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Outline

Transition: Single Variable to Comparing Groups

Univariate Analysis: What We've Covered

One variable measured on each sampling unit

- Population proportion: p (e.g., % of defective parts)
- Population mean: μ (e.g., average tensile strength)

Example: Measuring the Young's modulus of 50 steel specimens

Bivariate Analysis: What's Coming Next

Two variables measured on each sampling unit

- **Explanatory variable** (independent): Identifies which group/condition
- **Response variable** (dependent): The outcome we measure

Example: Comparing fatigue life (*response*) of specimens under two heat treatment processes (*explanatory*)

Intuition: *The explanatory variable tells us which experimental condition or population the observation belongs to, while the response variable is the measurement we're analyzing.*

Four Cases We'll Study

Categorical
Two distinct groups
Compare proportions

Quantitative
Two distinct groups
Compare means

Quantitative
Paired/Repeated
Compare paired means

Quantitative
Two groups
Compare variances

Key Decision

First: Is the response categorical or quantitative?
Second: Are the samples independent or paired?

Learning Objectives

Upon completion of this chapter, you should be able to:

1. Compare two population **proportions** using confidence intervals and hypothesis tests
2. Distinguish between **independent** and **paired** data when analyzing means
3. Compare two **independent means** with equal variances
4. Compare two **independent means** with unequal variances
5. Compare two **dependent (paired) means**
6. Compare two population **variances** using hypothesis tests

Difference of Two Independent Normal Variables

Theorem

Let $X \sim N(\mu_x, \sigma_x^2)$ *and* $Y \sim N(\mu_y, \sigma_y^2)$ *be independent.*
Then:

$$X - Y \sim N\left(\mu_x - \mu_y, \sigma_x^2 + \sigma_y^2\right)$$

Key Insights

- Mean of difference = difference of means
- Variance of difference = **sum** of variances (not difference!)
- Standard deviation: $\sqrt{\sigma_x^2 + \sigma_y^2}$

Applying the Central Limit Theorem

Sample Means

- If X and Y normal: $\bar{X}$ and $\bar{Y}$ are normal
- If sample sizes large: $\bar{X}$ and $\bar{Y}$ approximately normal (CLT)

Difference

$\bar{X} - \bar{Y}$ is approximately normal!

$\bar{X} \sim N(\mu_1, \sigma_1^2/n_1)$

Difference

$\bar{Y} \sim N(\mu_2, \sigma_2^2/n_2)$

$\bar{X} - \bar{Y}$
approximately normal

Intuition: *This theory enables us to build confidence intervals and hypothesis tests for differences!*

Example

Scenario:
Males and females were asked: If you received a $100 bill by mail, addressed to your neighbor but wrongly delivered to you, would you return it?

- Of 69 males sampled, 52 said "yes"
- Of 131 females sampled, 120 said "yes"

Research Question:
Do the data indicate that the proportions who said "yes" are **different** for males and females?

Let p_1 = proportion of males who would return it
Let p_2 = proportion of females who would return it

Point Estimate for $p_1 - p_2$

Definition

The point estimate for the difference between two population proportions is:

$$\hat{p}_1 - \hat{p}_2$$

Example

For the \$100 bill example:

$$\hat{p}_1 = \frac{52}{69} = 0.7536$$
$$\hat{p}_2 = \frac{120}{131} = 0.9160$$
$$\hat{p}_1 - \hat{p}_2 = 0.7536 - 0.9160 = -0.1624$$

Intuition: *Negative value suggests females have a higher proportion. But is this difference statistically significant?*

Sampling Distribution of $\hat{p}_1 - \hat{p}_2$

Conditions:
Need ALL four to be satisfied:

$$n_1\hat{p}_1 \geq 5, \quad n_1(1-\hat{p}_1) \geq 5, \quad n_2\hat{p}_2 \geq 5, \quad n_2(1-\hat{p}_2) \geq 5$$

Sampling Distribution:
If conditions satisfied and samples independent:

$$\hat{p}_1 - \hat{p}_2 \sim N\left(p_1 - p_2, \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\right)$$

Estimated Standard Error: Since we don't know p_1 and p_2:

$$SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Confidence Interval for $p_1 - p_2$

Formula:
The $(1-\alpha)100\%$ confidence interval for $p_1 - p_2$ is:

$$\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Example:[$100 Bill – 95% CI]

$$\begin{aligned} &-0.1624 \pm 1.96\sqrt{\frac{0.7536(0.2464)}{69} + \frac{0.9160(0.0840)}{131}} \\ &= -0.1624 \pm 0.1122 \\ &= (-0.2746, -0.0502) \end{aligned}$$

Interpretation:
We are 95% confident that the difference in proportions (males minus females) is between -0.27 and -0.05. Since both endpoints are negative, females appear more likely to return the money.

Hypothesis Test: The Null Value Always Zero

Previous Lesson

The null value could vary (e.g., $H_0\colon \mu = 100$)

This Lesson

When comparing two parameters, we use a null value of **0** ("no difference")

$$H_0\colon p_1 - p_2 = 0 \quad \Leftrightarrow \quad p_1 = p_2$$
$$H_0\colon \mu_1 - \mu_2 = 0 \quad \Leftrightarrow \quad \mu_1 = \mu_2$$

Intuition: *Zero difference means "the two populations are the same." This is the most common research question: Are these two groups different?*

Hypothesis Test for $p_1 - p_2$: Setup

Hypotheses

$$H_0: p_1 - p_2 = 0$$

Alternatives:

$$H_a: p_1 - p_2 \neq 0$$
$$H_a: p_1 - p_2 > 0$$
$$H_a: p_1 - p_2 < 0$$

Key Difference

Under H_0, we assume $p_1 = p_2 = p^*$ (some common value)

We estimate this common proportion using:

$$\hat{p}^* = \frac{x_1 + x_2}{n_1 + n_2}$$

Intuition: *This is called a **pooled estimate** because we combine data from both samples.*

Hypothesis Test for $p_1 - p_2$: Test Statistic

Test Statistic

$$z^* = \frac{\hat{p}_1 - \hat{p}_2 - 0}{\sqrt{\hat{p}^*(1-\hat{p}^*)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where $\hat{p}^* = \dfrac{x_1 + x_2}{n_1 + n_2}$

Example

$$\hat{p}^* = \frac{52 + 120}{69 + 131} = \frac{172}{200} = 0.86$$

$$z^* = \frac{0.7536 - 0.9160}{\sqrt{0.86(0.14)\left(\frac{1}{69} + \frac{1}{131}\right)}}$$

Hypothesis Test for $p_1 - p_2$: Conclusion

Example

- Test statistic: $z^* = -3.15$
- Two-sided test: p-value $= 2P(Z > |-3.15|) = 2(0.0008) = 0.0016$
- Significance level: $\alpha = 0.05$

Decision

Since p-value $= 0.0016 < 0.05$, we **reject** H_0.

Conclusion

There is sufficient evidence to conclude that the proportions of males and females who would return the $100 are **different** at the 5% significance level.

Intuition: *Notice the CI didn't contain 0, and we rejected H_0. These results are consistent!*

Independent vs. Dependent Samples

Independent Samples:
Samples are **independent** if the selection from one population has no relationship with the selection from the other population.

Dependent Samples:
Samples are **dependent (paired)** if each measurement in one sample is matched with a particular measurement in the other sample.

Rule of Thumb

- **One measurement per subject** $\Rightarrow$ Independent
- **Two measurements per subject** $\Rightarrow$ Paired
- Exception: Familial relationships (spouses, twins) $\Rightarrow$ Paired

Example

Question

We want to compare the gas mileage of two brands of gasoline (Brand A and Brand B).

Independent Design

- Randomly assign 12 cars to Brand A
- Randomly assign 12 different cars to Brand B
- Compare average mileage between groups

Paired Design

- Use 12 cars total
- Have each car use **both** Brand A and Brand B
- Compare differences in mileage *within* each car

Intuition: *Paired design controls for car-to-car variability and is usually more powerful!*

Two Cases for Independent Means

When comparing two independent means, we must consider the **population variances**:

Case 1: Equal Variances
$\sigma_1^2 = \sigma_2^2$
Use pooled t-test

Case 2: Unequal Variances
$\sigma_1^2 \neq \sigma_2^2$
Use unpooled t-test

Rule of Thumb

If sample sizes approximately equal and $0.5 \leq \frac{s_1}{s_2} \leq 1.5$, assume equal variances.

Pooled Standard Deviation

When to Use

When we can assume $\sigma_1^2 = \sigma_2^2$, we can pool information from both samples to estimate the common standard deviation.

Definition (Pooled Standard Deviation)

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Intuition: *This is a weighted average of the two sample variances. Larger samples get more weight!*

Degrees of Freedom

For the pooled test: $df = n_1 + n_2 - 2$

Confidence Interval: Pooled Variances

Assumptions

- Populations are independent
- Population variances are equal ($\sigma_1^2 = \sigma_2^2$)
- Each population is normal OR sample size is large

Formula

The $(1-\alpha)100\%$ confidence interval for $\mu_1 - \mu_2$ is:

$$\bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where $t_{\alpha/2}$ has $n_1 + n_2 - 2$ degrees of freedom.

Hypothesis Test: Pooled t-Test

Hypotheses

$$H_0\colon \mu_1 - \mu_2 = 0$$

Alternatives:

$$H_a\colon \mu_1 - \mu_2 \neq 0$$
$$H_a\colon \mu_1 - \mu_2 > 0$$
$$H_a\colon \mu_1 - \mu_2 < 0$$

Test Statistic

$$t^* = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

t^* follows a t-distribution with $df = n_1 + n_2 - 2$

Decision Rule

Compare p-value to α or compare t^* to critical value.

Example: Packing Machine

Scenario

A plant wants to test if a new machine packs cartons faster than the old machine. Times (in seconds) to pack 10 cartons were recorded:

New Machine: $n_1 = 10$, $\bar{x}_1 = 42.14$, $s_1 = 0.683$
Old Machine: $n_2 = 10$, $\bar{x}_2 = 43.23$, $s_2 = 0.750$

New Machine

42.1	41.3	42.4	43.2	41.8
41.0	41.8	42.8	42.3	42.7

Old Machine

42.7	43.8	42.5	43.1	44.0
43.6	43.3	43.5	41.7	44.1

Example: Packing Machines

Scenario

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Checking Assumptions

- Independent samples? Yes
- Normal distributions? Probability plots look good
- Equal variances? Check ratio: $\frac{s_1}{s_2} = \frac{0.683}{0.750} = 0.91$ Close to 1

Example: Packing Machines

Scenario

A plant wants to test if a new machine packs cartons faster than the old machine. Times (in seconds) to pack 10 cartons were recorded:

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Checking Assumptions

- Independent samples? Yes
- Normal distributions? Probability plots look good
- Equal variances? Check ratio: $\frac{s_1}{s_2} = \frac{0.683}{0.750} = 0.91$ Close to 1

Proceed with pooled t-test!

Example: Packing Machines – Hypothesis Test

Step 1: Hypotheses

$$H_0\colon \mu_1 - \mu_2 = 0 \quad \text{vs.} \quad H_a\colon \mu_1 - \mu_2 < 0$$

(Testing if new machine is faster, i.e., lower mean time)

Step 2: Calculate pooled standard deviation

$$s_p = \sqrt{\frac{(10-1)(0.683)^2 + (10-1)(0.750)^2}{10+10-2}} = \sqrt{\frac{9.261}{18}} = 0.717$$

Step 3: Test statistic

$$t^* = \frac{42.14 - 43.23}{0.717\sqrt{\frac{1}{10} + \frac{1}{10}}} = \frac{-1.09}{0.321} = -3.40$$

Example: Packing Machines – Conclusion

Step 4: Critical value and decision

- $\alpha = 0.05$, left-tailed test, $df = 18$
- Critical value: $t_{0.05,18} = -1.734$
- Rejection region: $t^* < -1.734$

Decision

Since $t^* = -3.40 < -1.734$, we **reject** H_0.

Conclusion

At the 5% significance level, there is sufficient evidence to conclude that the new machine packs cartons faster than the old machine.

99% Confidence Interval:

$$-1.09 \pm 2.878(0.321) = (-2.013, -0.167)$$

Unpooled (Separate Variances) Test

When to Use

When we **cannot** assume $\sigma_1^2 = \sigma_2^2$ (variances are unequal)

Test Statistic

$$t^* = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Degrees of Freedom

Complex formula (software calculates). Conservative approach: use $\min(n_1 - 1, n_2 - 1)$

$$df = \frac{(n_1 - 1)(n_2 - 1)}{(n_2 - 1)C^2 + (1 - C)^2(n_1 - 1)} \qquad \text{where } C = \frac{\frac{s_1^2}{n_1}}{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Confidence Interval

$$\bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Example: GPA – Test Results

Question: Do mean GPAs differ between sophomores and juniors? (Use $\alpha = 0.05$)

Sophomores

3.04	2.92	2.86	1.71	3.60
3.49	3.30	2.28	3.11	2.88
2.82	2.13	2.11	3.03	3.27
2.60	3.13			

Juniors

2.56	3.47	2.65	2.77	3.26
3.00	2.70	3.20	3.39	3.00
3.19	2.58	2.98		

Example: GPA by Class Level

Data

Independent random samples of sophomores and juniors:

Sophomores: $n_1 = 17$, $\bar{x}_1 = 2.840$, $s_1 = 0.520$
Juniors: $n_2 = 13$, $\bar{x}_2 = 2.981$, $s_2 = 0.309$

Check Variance Assumption

$$\frac{s_1}{s_2} = \frac{0.520}{0.309} = 1.68$$

Standard deviations are quite different and sample sizes are small $\Rightarrow$ Use **unpooled test**

Example: GPA – Test Results

Question: Do mean GPAs differ between sophomores and juniors? (Use $\alpha = 0.05$)

Hypotheses:

$$H_0\colon \mu_1 - \mu_2 = 0 \quad \text{vs.} \quad H_a\colon \mu_1 - \mu_2 \neq 0$$

Software Output:

- Test statistic: $t^* = -0.92$
- Degrees of freedom: $df = 26$
- p-value $= 0.364$
- 95% CI: $(-0.454, 0.173)$

Conclusion

Since p-value $= 0.364 > 0.05$, we **fail to reject** H_0. There is insufficient evidence to conclude that mean GPAs differ between sophomores and juniors.

The Paired Data Approach

Key Insight

When data are paired, we can "condense" two measurements into one by taking the **difference**!

Example

Weight loss study: Measure weight before and after diet

- Instead of two variables (before, after)
- Create one variable: difference = before − after
- The difference is the weight lost!

Intuition: *Once we compute differences, we're back to the **one-sample** case! Apply one-sample t-methods to the differences.*

Notation for Paired Data

Suppose we have n pairs of observations:

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$

Definition (Differences)

$$d_i = x_i - y_i \quad \text{for } i = 1, 2, \ldots, n$$

Definition (Sample Mean of Differences)

$$\bar{d} = \frac{1}{n} \sum_{i=1}^{n} d_i$$

Definition (Sample Standard Deviation of Differences)

s_d = standard deviation of the d_i values

Confidence Interval for μ_d

Assumptions

- Differences come from a normal distribution, OR
- Sample size is large ($n > 30$)

Formula

The $(1-\alpha)100\%$ confidence interval for μ_d is:

$$\bar{d} \pm t_{\alpha/2} \frac{s_d}{\sqrt{n}}$$

where $t_{\alpha/2}$ has $n-1$ degrees of freedom.

Intuition: *This is exactly the same as the one-sample t-interval, applied to the differences!*

Hypothesis Test for μ_d

Hypotheses

$$H_0\colon \mu_d = 0$$

Alternatives:

$$H_a\colon \mu_d \neq 0$$
$$H_a\colon \mu_d > 0$$
$$H_a\colon \mu_d < 0$$

Test Statistic

$$t^* = \frac{\bar{d} - 0}{s_d/\sqrt{n}}$$

t^* follows a t-distribution with $df = n - 1$

Conditions

Differences are approximately normal OR $n > 30$

Example: Zinc Concentrations in Water

Data

Ten locations measured zinc concentration in bottom water and surface water:

Location	1	2	3	4	5	6	7	8	9	10
Bottom	.430	.266	.567	.531	.707	.716	.651	.589	.469	.723
Surface	.415	.238	.390	.410	.605	.609	.632	.523	.411	.612
Difference	.015	.028	.177	.121	.102	.107	.019	.066	.058	.111

Summary of differences: $n = 10$, $\bar{d} = 0.0804$, $s_d = 0.0523$

Example: Zinc – Confidence Interval

Question: Does the data suggest the true average concentration in bottom water differs from surface water? Use 95% CI.

Check assumptions:

- Small sample ($n = 10$), so check normality of differences
- Normal probability plot shows no clear violations

Calculate CI:

$$\bar{d} \pm t_{0.025,9} \cdot \frac{s_d}{\sqrt{n}}$$
$$0.0804 \pm 2.262 \cdot \frac{0.0523}{\sqrt{10}}$$
$$0.0804 \pm 0.0374$$
$$= (0.043, 0.118)$$

Interpretation: We are 95% confident that bottom water has, on average, 0.043 to 0.118 higher zinc concentration than surface water. Since 0 is not in the interval, the means are significantly different.

Example: Zinc – Hypothesis Test

Question: Does bottom water have higher concentration than surface water? Use $\alpha = 0.05$.

Hypotheses:

$$H_0\colon \mu_d = 0 \quad \text{vs.} \quad H_a\colon \mu_d > 0$$

Test statistic:

$$t^* = \frac{0.0804}{0.0523/\sqrt{10}} = \frac{0.0804}{0.0165} = 4.86$$

Critical value: $t_{0.05,9} = 1.833$
Rejection region: $t^* > 1.833$

Conclusion

Since $t^* = 4.86 > 1.833$, we reject H_0. There is strong evidence that bottom water has higher zinc concentration than surface water.

Why Pairing Matters

Important Note

For the zinc concentration problem:

- If you mistakenly use a 2-sample t-test (treating as independent)
- You will **not** be able to reject the null hypothesis!

Intuition: *Why? Pairing removes location-to-location variability, making the test more powerful. Each location serves as its own control.*

Design Lesson

When possible, design experiments with pairing! It's more efficient and leads to:

- Smaller standard errors
- Narrower confidence intervals
- Greater power to detect differences

Why Compare Variances?

Practical Reasons

- **Quality control:** Choose the process with smaller variability
- **Consistency:** Compare reliability of different methods
- **Assumption checking:** Verify equal variances assumption for pooled t-test

Example Scenarios

- Two manufacturing processes – which has more consistent output?
- Two teaching methods – which produces more uniform learning?
- Two measurement devices – which is more precise?

F-Test for Comparing Two Variances

Hypotheses

Null: $H_0\colon \dfrac{\sigma_1^2}{\sigma_2^2} = 1$ (equivalently, $\sigma_1^2 = \sigma_2^2$)

Alternatives:

$$H_a\colon \frac{\sigma_1^2}{\sigma_2^2} \neq 1$$

$$H_a\colon \frac{\sigma_1^2}{\sigma_2^2} > 1$$

$$H_a\colon \frac{\sigma_1^2}{\sigma_2^2} < 1$$

Critical Assumption

The F-test requires that **both populations are normally distributed**. This is NOT negotiable!

F-Test Statistic and Alternatives

Test Statistic

$$F = \frac{s_1^2}{s_2^2}$$

Under H_0, F follows an F-distribution with $df_1 = n_1 - 1$ and $df_2 = n_2 - 1$

Alternative Tests (More Robust)

When normality is questionable:

- **Bonett's test:** Only assumes data is quantitative
- **Levene's test:** Best if data heavily skewed and both $n_i < 20$

Warning

Unlike means, CLT does NOT apply to variances! Large sample size does not excuse non-normality for the F-test.

Example: Packing Machine Variances

Question

For the packing machine data, is it reasonable to assume equal population variances?

New Machine: $n_1 = 10$, $s_1 = 0.683$
Old Machine: $n_2 = 10$, $s_2 = 0.750$

Check normality: Probability plots for both samples show no violations

Test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{(0.683)^2}{(0.750)^2} = \frac{0.467}{0.562} = 0.83$$

Since F-tables are usually read with the larger variance in the numerator, we also write

$$F^* = \frac{0.562}{0.467} = 1.204$$

with $df_1 = 9$ and $df_2 = 9$.

Example: Packing Machine Variances (contd.)

For a two-sided test at $\alpha = 0.05$, the upper critical value is

$$F_{0.025,9,9} \approx 4.03$$

The corresponding lower critical value is the reciprocal:

$$\frac{1}{4.03} \approx 0.248$$

So the rejection region is

$$F < 0.248 \quad \text{or} \quad F > 4.03$$

Since

$$0.248 < \mathbf{1.204} < 4.03,$$

we do not reject H_0.

Example: Packing Machine – Conclusion

Decision

Since p-value $= 0.787 > 0.05$, we **fail to reject** H_0.

Conclusion

There is insufficient evidence to conclude that the population variances are different. It is reasonable to assume equal variances.

Intuition: *This confirms our earlier decision to use the* ***pooled*** *t-test when comparing the means!*

Reminder

Always check this assumption *before* deciding between pooled and unpooled tests!

Summary: Test Statistics

Comparison	Test Statistic	Distribution
Two proportions	$z^* = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}^*(1-\hat{p}^*)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$	Standard normal
Pooled means	$t^* = \dfrac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$	$t_{n_1+n_2-2}$
Unpooled means	$t^* = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	t_{df} (complex)
Paired means	$t^* = \dfrac{\bar{d}}{s_d/\sqrt{n}}$	t_{n-1}
Two variances	$F = \dfrac{s_1^2}{s_2^2}$	F_{n_1-1,n_2-1}

Summary: Key Concepts

1. **Always start with research question:**
   - What are we comparing? (proportions, means, variances)
   - Are samples independent or paired?
2. **Check assumptions before proceeding:**
   - Sample size conditions (for proportions)
   - Normality (probability plots for small samples)
   - Equal variances (rule of thumb or F-test)
3. **Null hypothesis is always "no difference":**

$$H_0 \colon p_1 - p_2 = 0, \quad H_0 \colon \mu_1 - \mu_2 = 0, \quad H_0 \colon \mu_d = 0$$

4. **Confidence intervals complement hypothesis tests:**
   - If CI contains 0 $\Rightarrow$ fail to reject H_0
   - If CI doesn't contain 0 $\Rightarrow$ reject H_0